Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
calculate the mean of a set of numbers.			
find the median of a set of numbers.			
find the mode of a set of numbers.			
use the basic definition of probability.			

ⓑ *After looking at the checklist, do you think you are well prepared for the next section? Why or why not?*

5.6 Ratios and Rate

Learning Objectives

By the end of this section, you will be able to:

- Write a ratio as a fraction
- Write a rate as a fraction
- Find unit rates
- Find unit price
- Translate phrases to expressions with fractions

☑ **BE PREPARED** 5.16 Before you get started, take this readiness quiz.

Simplify: $\frac{16}{24}$.

If you missed this problem, review Example 4.19.

☑ **BE PREPARED** 5.17 Divide: $2.76 \div 11.5$.

If you missed this problem, review Example 5.19.

☑ **BE PREPARED** 5.18 Simplify: $\frac{1\frac{1}{2}}{2\frac{3}{4}}$.

If you missed this problem, review Example 4.43.

Write a Ratio as a Fraction

When you apply for a mortgage, the loan officer will compare your total debt to your total income to decide if you qualify for the loan. This comparison is called the debt-to-income ratio. A **ratio** compares two quantities that are measured with the same unit. If we compare a and b, the ratio is written as a to b, $\frac{a}{b}$, or $a{:}b$.

Ratios

A **ratio** compares two numbers or two quantities that are measured with the same unit. The ratio of a to b is written a to b, $\frac{a}{b}$, or $a{:}b$.

In this section, we will use the fraction notation. When a ratio is written in fraction form, the fraction should be simplified. If it is an improper fraction, we do not change it to a mixed number. Because a ratio compares two quantities, we would leave a ratio as $\frac{4}{1}$ instead of simplifying it to 4 so that we can see the two parts of the ratio.

EXAMPLE 5.58

Write each ratio as a fraction: ⓐ 15 to 27 ⓑ 45 to 18.

Solution

ⓐ

	15 to 27
Write as a fraction with the first number in the numerator and the second in the denominator.	$\frac{15}{27}$
Simplify the fraction.	$\frac{5}{9}$

ⓑ

	45 to 18
Write as a fraction with the first number in the numerator and the second in the denominator.	$\frac{45}{18}$
Simplify.	$\frac{5}{2}$

We leave the ratio in ⓑ as an improper fraction.

TRY IT 5.115 Write each ratio as a fraction: ⓐ 21 to 56 ⓑ 48 to 32.

TRY IT 5.116 Write each ratio as a fraction: ⓐ 27 to 72 ⓑ 51 to 34.

Ratios Involving Decimals

We will often work with ratios of decimals, especially when we have ratios involving money. In these cases, we can eliminate the decimals by using the Equivalent Fractions Property to convert the ratio to a fraction with whole numbers in the numerator and denominator.

For example, consider the ratio 0.8 to 0.05. We can write it as a fraction with decimals and then multiply the numerator and denominator by 100 to eliminate the decimals.

$$\frac{0.8}{0.05}$$
$$\frac{(0.8)100}{(0.05)100}$$
$$\frac{80}{5}$$

Do you see a shortcut to find the equivalent fraction? Notice that $0.8 = \frac{8}{10}$ and $0.05 = \frac{5}{100}$. The least common denominator of $\frac{8}{10}$ and $\frac{5}{100}$ is 100. By multiplying the numerator and denominator of $\frac{0.8}{0.05}$ by 100, we 'moved' the decimal two places to the right to get the equivalent fraction with no decimals. Now that we understand the math behind the process, we can find the fraction with no decimals like this:

	$\frac{0.80}{0.05}$
"Move" the decimal 2 places.	$\frac{80}{5}$
Simplify.	$\frac{16}{1}$

You do not have to write out every step when you multiply the numerator and denominator by powers of ten. As long as you move both decimal places the same number of places, the ratio will remain the same.

EXAMPLE 5.59

Write each ratio as a fraction of whole numbers:

ⓐ 4.8 to 11.2 ⓑ 2.7 to 0.54

Solution

ⓐ 4.8 to 11.2

Write as a fraction.	$\frac{4.8}{11.2}$
Rewrite as an equivalent fraction without decimals, by moving both decimal points 1 place to the right.	$\frac{48}{112}$
Simplify.	$\frac{3}{7}$

So 4.8 to 11.2 is equivalent to $\frac{3}{7}$.

ⓑ
The numerator has one decimal place and the denominator has 2. To clear both decimals we need to move the decimal 2 places to the right.
2.7 to 0.54

Write as a fraction.	$\frac{2.7}{0.54}$
Move both decimals right two places.	$\frac{270}{54}$
Simplify.	$\frac{5}{1}$

So 2.7 to 0.54 is equivalent to $\frac{5}{1}$.

TRY IT 5.117 Write each ratio as a fraction: ⓐ 4.6 to 11.5 ⓑ 2.3 to 0.69.

TRY IT 5.118 Write each ratio as a fraction: ⓐ 3.4 to 15.3 ⓑ 3.4 to 0.68.

Some ratios compare two mixed numbers. Remember that to divide mixed numbers, you first rewrite them as improper fractions.

EXAMPLE 5.60

Write the ratio of $1\frac{1}{4}$ to $2\frac{3}{8}$ as a fraction.

Solution

	$1\frac{1}{4}$ to $2\frac{3}{8}$
Write as a fraction.	$\frac{1\frac{1}{4}}{2\frac{3}{8}}$
Convert the numerator and denominator to improper fractions.	$\frac{\frac{5}{4}}{\frac{19}{8}}$
Rewrite as a division of fractions.	$\frac{5}{4} \div \frac{19}{8}$
Invert the divisor and multiply.	$\frac{5}{4} \cdot \frac{8}{19}$
Simplify.	$\frac{10}{19}$

TRY IT 5.119 Write each ratio as a fraction: $1\frac{3}{4}$ to $2\frac{5}{8}$.

TRY IT 5.120 Write each ratio as a fraction: $1\frac{1}{8}$ to $2\frac{3}{4}$.

Applications of Ratios

One real-world application of ratios that affects many people involves measuring cholesterol in blood. The ratio of total cholesterol to HDL cholesterol is one way doctors assess a person's overall health. A ratio of less than 5 to 1 is considered good.

EXAMPLE 5.61

Hector's total cholesterol is 249 mg/dl and his HDL cholesterol is 39 mg/dl. (a) Find the ratio of his total cholesterol to his HDL cholesterol. (b) Assuming that a ratio less than 5 to 1 is considered good, what would you suggest to Hector?

Solution

(a) First, write the words that express the ratio. We want to know the ratio of Hector's total cholesterol to his HDL cholesterol.

Write as a fraction.	$\frac{\text{total cholesterol}}{\text{HDL cholesterol}}$
Substitute the values.	$\frac{249}{39}$
Simplify.	$\frac{83}{13}$

(b) Is Hector's cholesterol ratio ok? If we divide 83 by 13 we obtain approximately 6.4, so $\frac{83}{13} \approx \frac{6.4}{1}$. Hector's cholesterol ratio is high! Hector should either lower his total cholesterol or raise his HDL cholesterol.

TRY IT 5.121 Find the patient's ratio of total cholesterol to HDL cholesterol using the given information.

Total cholesterol is 185 mg/dL and HDL cholesterol is 40 mg/dL.

TRY IT 5.122 Find the patient's ratio of total cholesterol to HDL cholesterol using the given information.

Total cholesterol is 204 mg/dL and HDL cholesterol is 38 mg/dL.

Ratios of Two Measurements in Different Units

To find the ratio of two measurements, we must make sure the quantities have been measured with the same unit. If the measurements are not in the same units, we must first convert them to the same units.

We know that to simplify a fraction, we divide out common factors. Similarly in a ratio of measurements, we divide out the common unit.

EXAMPLE 5.62

The Americans with Disabilities Act (ADA) Guidelines for wheel chair ramps require a maximum vertical rise of 1 inch for every 1 foot of horizontal run. What is the ratio of the rise to the run?

Solution

In a ratio, the measurements must be in the same units. We can change feet to inches, or inches to feet. It is usually easier to convert to the smaller unit, since this avoids introducing more fractions into the problem.

Write the words that express the ratio.

	Ratio of the rise to the run
Write the ratio as a fraction.	$\frac{\text{rise}}{\text{run}}$
Substitute in the given values.	$\frac{1 \text{ inch}}{1 \text{ foot}}$
Convert 1 foot to inches.	$\frac{1 \text{ inch}}{12 \text{ inches}}$
Simplify, dividing out common factors and units.	$\frac{1}{12}$

So the ratio of rise to run is 1 to 12. This means that the ramp should rise 1 inch for every 12 inches of horizontal run to comply with the guidelines.

TRY IT 5.123 Find the ratio of the first length to the second length: 32 inches to 1 foot.

TRY IT 5.124 Find the ratio of the first length to the second length: 1 foot to 54 inches.

Write a Rate as a Fraction

Frequently we want to compare two different types of measurements, such as miles to gallons. To make this comparison, we use a **rate**. Examples of rates are 120 miles in 2 hours, 160 words in 4 minutes, and $5 dollars per 64 ounces.

Rate

A **rate** compares two quantities of different units. A rate is usually written as a fraction.

When writing a fraction as a rate, we put the first given amount with its units in the numerator and the second amount with its units in the denominator. When rates are simplified, the units remain in the numerator and denominator.

EXAMPLE 5.63

Bob drove his car 525 miles in 9 hours. Write this rate as a fraction.

Solution

	525 miles in 9 hours
Write as a fraction, with 525 miles in the numerator and 9 hours in the denominator.	$\frac{525 \text{ miles}}{9 \text{ hours}}$
	$\frac{175 \text{ miles}}{3 \text{ hours}}$

So 525 miles in 9 hours is equivalent to $\frac{175 \text{ miles}}{3 \text{ hours}}$.

TRY IT 5.125 Write the rate as a fraction: 492 miles in 8 hours.

TRY IT 5.126 Write the rate as a fraction: 242 miles in 6 hours.

Find Unit Rates

In the last example, we calculated that Bob was driving at a rate of $\frac{175 \text{ miles}}{3 \text{ hours}}$. This tells us that every three hours, Bob will travel 175 miles. This is correct, but not very useful. We usually want the rate to reflect the number of miles in one hour. A rate that has a denominator of 1 unit is referred to as a **unit rate**.

Unit Rate

A **unit rate** is a rate with denominator of 1 unit.

Unit rates are very common in our lives. For example, when we say that we are driving at a speed of 68 miles per hour we mean that we travel 68 miles in 1 hour. We would write this rate as 68 miles/hour (read 68 miles per hour). The common abbreviation for this is 68 mph. Note that when no number is written before a unit, it is assumed to be 1.

So 68 miles/hour really means 68 miles/1 hour.

Two rates we often use when driving can be written in different forms, as shown:

Example	Rate	Write	Abbreviate	Read
68 miles in 1 hour	$\frac{68 \text{ miles}}{1 \text{ hour}}$	68 miles/hour	68 mph	68 miles per hour
36 miles to 1 gallon	$\frac{36 \text{ miles}}{1 \text{ gallon}}$	36 miles/gallon	36 mpg	36 miles per gallon

Another example of unit rate that you may already know about is hourly pay rate. It is usually expressed as the amount of money earned for one hour of work. For example, if you are paid $12.50 for each hour you work, you could write that your hourly (unit) pay rate is $12.50/hour (read $12.50 per hour.)

To convert a rate to a unit rate, we divide the numerator by the denominator. This gives us a denominator of 1.

EXAMPLE 5.64

Anita was paid $384 last week for working 32 hours. What is Anita's hourly pay rate?

Solution

Start with a rate of dollars to hours. Then divide.	$384 last week for 32 hours
Write as a rate.	$\frac{\$384}{32 \text{ hours}}$
Divide the numerator by the denominator.	$\frac{\$12}{1 \text{ hour}}$
Rewrite as a rate.	$12/hour

Anita's hourly pay rate is $12 per hour.

TRY IT 5.127 Find the unit rate: $630 for 35 hours.

TRY IT 5.128 Find the unit rate: $684 for 36 hours.

EXAMPLE 5.65

Sven drives his car 455 miles, using 14 gallons of gasoline. How many miles per gallon does his car get?

Solution

Start with a rate of miles to gallons. Then divide.

	455 miles to 14 gallons of gas
Write as a rate.	$\frac{455 \text{ miles}}{14 \text{ gallons}}$
Divide 455 by 14 to get the unit rate.	$\frac{32.5 \text{ miles}}{1 \text{ gallon}}$

Sven's car gets 32.5 miles/gallon, or 32.5 mpg.

TRY IT 5.129 Find the unit rate: 423 miles to 18 gallons of gas.

TRY IT 5.130 Find the unit rate: 406 miles to 14.5 gallons of gas.

Find Unit Price

Sometimes we buy common household items 'in bulk', where several items are packaged together and sold for one price. To compare the prices of different sized packages, we need to find the unit price. To find the unit price, divide the total price by the number of items. A **unit price** is a unit rate for one item.

Unit price

A **unit price** is a unit rate that gives the price of one item.

EXAMPLE 5.66

The grocery store charges $3.99 for a case of 24 bottles of water. What is the unit price?

Solution

What are we asked to find? We are asked to find the unit price, which is the price per bottle.

Write as a rate.	$\frac{\$3.99}{24 \text{ bottles}}$
Divide to find the unit price.	$\frac{\$0.16625}{1 \text{ bottle}}$
Round the result to the nearest penny.	$\frac{\$0.17}{1 \text{ bottle}}$

The unit price is approximately $0.17 per bottle. Each bottle costs about $0.17.

TRY IT 5.131 Find the unit price. Round your answer to the nearest cent if necessary.

24-pack of juice boxes for $6.99

TRY IT 5.132 Find the unit price. Round your answer to the nearest cent if necessary.

24-pack of bottles of ice tea for $12.72

Unit prices are very useful if you comparison shop. The *better buy* is the item with the lower unit price. Most grocery stores list the unit price of each item on the shelves.

EXAMPLE 5.67

Paul is shopping for laundry detergent. At the grocery store, the liquid detergent is priced at $14.99 for 64 loads of laundry and the same brand of powder detergent is priced at $15.99 for 80 loads.

Which detergent has the lowest cost per load?

Solution

To compare the prices, we first find the unit price for each type of detergent.

	Liquid	Powder
Write as a rate.	$\frac{\$14.99}{64 \text{ loads}}$	$\frac{\$15.99}{80 \text{ loads}}$
Find the unit price.	$\frac{\$0.234\ldots}{1 \text{ load}}$	$\frac{\$0.199\ldots}{1 \text{ load}}$
Round to the nearest cent.	$0.23/load (23 cents per load.)	$0.20/load (20 cents per load)

Now we compare the unit prices. The unit price of the liquid detergent is about $0.23 per load and the unit price of the powder detergent is about $0.20 per load. The powder is the better buy.

TRY IT 5.133 Find each unit price and then determine the better buy. Round to the nearest cent if necessary.

Brand A Storage Bags, $4.59 for 40 count, or Brand B Storage Bags, $3.99 for 30 count

TRY IT 5.134 Find each unit price and then determine the better buy. Round to the nearest cent if necessary.

Brand C Chicken Noodle Soup, $1.89 for 26 ounces, or Brand D Chicken Noodle Soup, $0.95 for 10.75 ounces

Notice in Example 5.67 that we rounded the unit price to the nearest cent. Sometimes we may need to carry the division to one more place to see the difference between the unit prices.

Translate Phrases to Expressions with Fractions

Have you noticed that the examples in this section used the comparison words *ratio of, to, per, in, for, on*, and *from*? When you translate phrases that include these words, you should think either ratio or rate. If the units measure the same quantity (length, time, etc.), you have a ratio. If the units are different, you have a rate. In both cases, you write a fraction.

EXAMPLE 5.68

Translate the word phrase into an algebraic expression:

ⓐ 427 miles per h hours ⓑ x students to 3 teachers ⓒ y dollars for 18 hours

Solution

ⓐ

	427 miles per h hours
Write as a rate.	$\frac{427 \text{ miles}}{h \text{ hours}}$

ⓑ

	x students to 3 teachers
Write as a rate.	$\frac{x \text{ students}}{3 \text{ teachers}}$

ⓒ

	y dollars for 18 hours
Write as a rate.	$\frac{\$y}{18 \text{ hours}}$

TRY IT 5.135 Translate the word phrase into an algebraic expression.

ⓐ 689 miles per h hours ⓑ y parents to 22 students ⓒ d dollars for 9 minutes

TRY IT 5.136 Translate the word phrase into an algebraic expression.

ⓐ m miles per 9 hours ⓑ x students to 8 buses ⓒ y dollars for 40 hours

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Ratios (https://www.openstax.org/l/24ratios)

Write Ratios as a Simplified Fractions Involving Decimals and Fractions (https://www.openstax.org/l/24ratiosimpfrac)
Write a Ratio as a Simplified Fraction (https://www.openstax.org/l/24ratiosimp)
Rates and Unit Rates (https://www.openstax.org/l/24rates)
Unit Rate for Cell Phone Plan (https://www.openstax.org/l/24unitrate)

SECTION 5.6 EXERCISES

Practice Makes Perfect

Write a Ratio as a Fraction

In the following exercises, write each ratio as a fraction.

403. 20 to 36

404. 20 to 32

405. 42 to 48

406. 45 to 54

407. 49 to 21

408. 56 to 16

409. 84 to 36

410. 6.4 to 0.8

411. 0.56 to 2.8

412. 1.26 to 4.2

413. $1\frac{2}{3}$ to $2\frac{5}{6}$

414. $1\frac{3}{4}$ to $2\frac{5}{8}$

415. $4\frac{1}{6}$ to $3\frac{1}{3}$

416. $5\frac{3}{5}$ to $3\frac{3}{5}$

417. \$18 to \$63

418. \$16 to \$72

419. \$1.21 to \$0.44

420. \$1.38 to \$0.69

421. 28 ounces to 84 ounces

422. 32 ounces to 128 ounces

423. 12 feet to 46 feet

424. 15 feet to 57 feet

425. 246 milligrams to 45 milligrams

426. 304 milligrams to 48 milligrams

427. total cholesterol of 175 to HDL cholesterol of 45

428. total cholesterol of 215 to HDL cholesterol of 55

429. 27 inches to 1 foot

430. 28 inches to 1 foot

Write a Rate as a Fraction

In the following exercises, write each rate as a fraction.

431. 140 calories per 12 ounces

432. 180 calories per 16 ounces

433. 8.2 pounds per 3 square inches

434. 9.5 pounds per 4 square inches

435. 488 miles in 7 hours

436. 527 miles in 9 hours

437. \$595 for 40 hours

438. \$798 for 40 hours

Find Unit Rates

In the following exercises, find the unit rate. Round to two decimal places, if necessary.

439. 140 calories per 12 ounces

440. 180 calories per 16 ounces

441. 8.2 pounds per 3 square inches

442. 9.5 pounds per 4 square inches

443. 488 miles in 7 hours

444. 527 miles in 9 hours

445. \$595 for 40 hours

446. \$798 for 40 hours

447. 576 miles on 18 gallons of gas

448. 435 miles on 15 gallons of gas

449. 43 pounds in 16 weeks

450. 57 pounds in 24 weeks

451. 46 beats in 0.5 minute

452. 54 beats in 0.5 minute

453. The bindery at a printing plant assembles 96,000 magazines in 12 hours. How many magazines are assembled in one hour?

454. The pressroom at a printing plant prints 540,000 sections in 12 hours. How many sections are printed per hour?

Find Unit Price

In the following exercises, find the unit price. Round to the nearest cent.

455. Soap bars at 8 for $8.69

456. Soap bars at 4 for $3.39

457. Women's sports socks at 6 pairs for $7.99

458. Men's dress socks at 3 pairs for $8.49

459. Snack packs of cookies at 12 for $5.79

460. Granola bars at 5 for $3.69

461. CD-RW discs at 25 for $14.99

462. CDs at 50 for $4.49

463. The grocery store has a special on macaroni and cheese. The price is $3.87 for 3 boxes. How much does each box cost?

464. The pet store has a special on cat food. The price is $4.32 for 12 cans. How much does each can cost?

In the following exercises, find each unit price and then identify the better buy. Round to three decimal places.

465. Mouthwash, 50.7-ounce size for $6.99 or 33.8-ounce size for $4.79

466. Toothpaste, 6 ounce size for $3.19 or 7.8-ounce size for $5.19

467. Breakfast cereal, 18 ounces for $3.99 or 14 ounces for $3.29

468. Breakfast Cereal, 10.7 ounces for $2.69 or 14.8 ounces for $3.69

469. Ketchup, 40-ounce regular bottle for $2.99 or 64-ounce squeeze bottle for $4.39

470. Mayonnaise 15-ounce regular bottle for $3.49 or 22-ounce squeeze bottle for $4.99

471. Cheese $6.49 for 1 lb. block or $3.39 for $\frac{1}{2}$ lb. block

472. Candy $10.99 for a 1 lb. bag or $2.89 for $\frac{1}{4}$ lb. of loose candy

Translate Phrases to Expressions with Fractions

In the following exercises, translate the English phrase into an algebraic expression.

473. 793 miles per p hours

474. 78 feet per r seconds

475. $3 for 0.5 lbs.

476. j beats in 0.5 minutes

477. 105 calories in x ounces

478. 400 minutes for m dollars

479. the ratio of y and $5x$

480. the ratio of $12x$ and y